

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jun 2024 P2HR Worked Solutions

Jun 2024 | P2HR

33 questions ■ question image followed by worked solution

PREPARED BY

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION**Q#: 1****Marks: 3****TOPIC: Prime Factors, HCF & LCM**

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- 1** Write 1400 as a product of powers of its prime factors.
Show your working clearly.

(Total for Question 1 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 1

Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

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WORKED SOLUTION

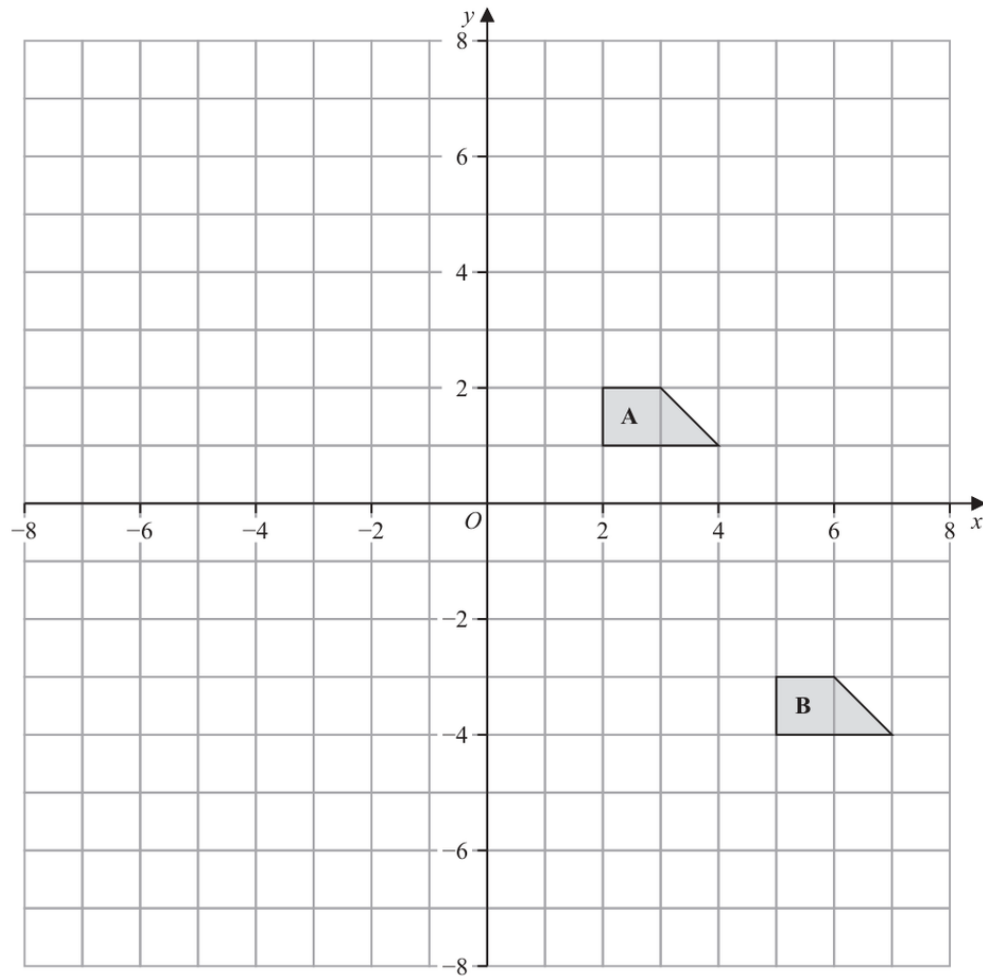
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 2****Marks: 4****TOPIC: Transformations**

Jun 2024 P2HR - Jun 2024 | P2HR

2

- (a) Describe fully the single transformation that maps shape **A** onto shape **B**

.....

 (2)

- (b) On the grid above, rotate shape **A** 180° about $(-1, 0)$
 Label your shape **C**

(2)

(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION

Q#: 2

Marks: 4

TOPIC: **Transformations**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 3****Marks: 2****TOPIC: Statistics Toolkit**

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- 3** Here is a list of four numbers written in ascending order of size

 $x \quad x \quad y \quad 15$

where x and y are integers.

The numbers have

a median of 12.5

a range of 4

Find the value of x and the value of y

 $x = \dots\dots\dots$ $y = \dots\dots\dots$

(Total for Question 3 is 2 marks)

PAST PAPER SOLUTION

Q#: 3

Marks: 2

TOPIC: **Statistics Toolkit**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 4****Marks: 5****TOPIC: Set Notation & Venn Diagrams**

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- 4 $\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
 $A = \{\text{factors of } 6\}$
 $B = \{\text{prime numbers}\}$

(a) List the members of the set

(i) $A \cup B$
(1)(ii) A'
(1)Harpreet states that $A \cap B = \emptyset$

Harpreet is incorrect.

(b) Explain why.

.....
(1) C is a set with 4 members such that

the set $A \cap C$ has 2 members
 the set $B \cap C$ has 2 members

Set $A \cap C$ and set $B \cap C$ have no members in common.(c) List the 4 members of set C
(2)**(Total for Question 4 is 5 marks)**

PAST PAPER SOLUTION

Q#: 4

Marks: 5

TOPIC: **Set Notation & Venn Diagrams**

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PAST PAPER QUESTION**Q#: 5****Marks: 4****TOPIC: Circles, Arcs & Sectors**

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- 5 The diagram shows the design for a badge, which will be made using wire.
The design is a circle inside a square $ABCD$

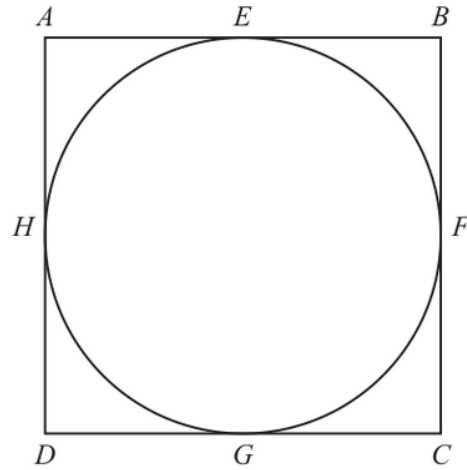


Diagram **NOT**
accurately drawn

The circle touches the square at the points E , F , G and H

The area of the square is 81 cm^2

Calculate the total length of wire that will be needed to make the square and the circle.
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION

Q#: 5

Marks: 4

TOPIC: **Circles, Arcs & Sectors**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 6****Marks: 8****TOPIC: Solving Linear Equations**

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6 (a) Solve $\frac{2f}{3} = 4f - 17$

Show clear algebraic working.

$$f = \dots\dots\dots$$

(3)

(b) Simplify $(e + 12)^0$ where $e > 0$

$$\dots\dots\dots$$

(1)

(c) Simplify fully $\frac{12a^4h^6}{4ah^2}$

$$\dots\dots\dots$$

(2)

(d) Factorise fully $20x^5y + 12x^3y^4$

$$\dots\dots\dots$$

(2)

(Total for Question 6 is 8 marks)

PAST PAPER SOLUTION

Q#: 6

Marks: 8

TOPIC: **Solving Linear Equations**

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PAST PAPER QUESTION**Q#: 7****Marks: 2****TOPIC: Algebraic Roots & Indices**

Jun 2024 P2HR - Jun 2024 | P2HR

7 $\frac{3^{-2} \times 3^5}{3^{10}} = 3^n$

Find the value of n $n = \dots\dots\dots$

(Total for Question 7 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 7

Marks: 2

TOPIC: **Algebraic Roots & Indices**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 8****Marks: 3**TOPIC: **Percentages**

Jun 2024 P2HR - Jun 2024 | P2HR

8 In a sale, all normal prices are reduced by 17%

The sale price of a fridge is 6225 rupees.

Work out the normal price of the fridge.

..... rupees

(Total for Question 8 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 8

Marks: 3

TOPIC: Percentages

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 9****Marks: 4****TOPIC: Powers, Roots & Standard Form**

Jun 2024 P2HR - Jun 2024 | P2HR

- 9 (a) Write 6.04×10^5 as an ordinary number.

.....
(1)

- (b) Write 0.000 07 in standard form.

.....
(1)

- (c) Work out $\frac{7.6 \times 10^{10}}{4 \times 10^5 - 2 \times 10^4}$

Give your answer in standard form.

.....
(2)

(Total for Question 9 is 4 marks)

PAST PAPER SOLUTION

Q#: 9

Marks: 4

TOPIC: **Powers, Roots & Standard Form**

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WORKED SOLUTION

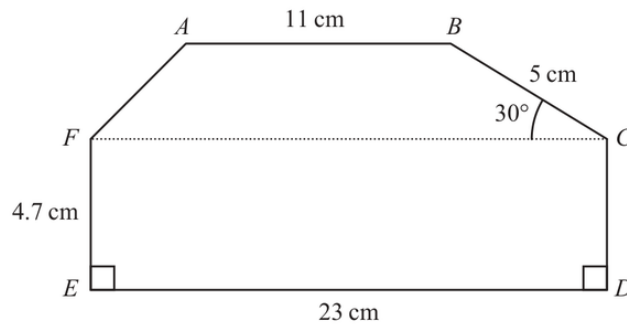
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 10****Marks: 5****TOPIC: Area & Perimeter**

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10 The diagram shows a hexagon $ABCDEF$ Diagram **NOT**
accurately drawnAngle $BCF = 30^\circ$ AB , FC and ED are parallel.Calculate the area of $ABCDEF$

Show your working clearly.

..... cm^2

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION

Q#: 10

Marks: 5

TOPIC: **Area & Perimeter**

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EA

PAST PAPER QUESTION

Q#: 11 Marks: 7

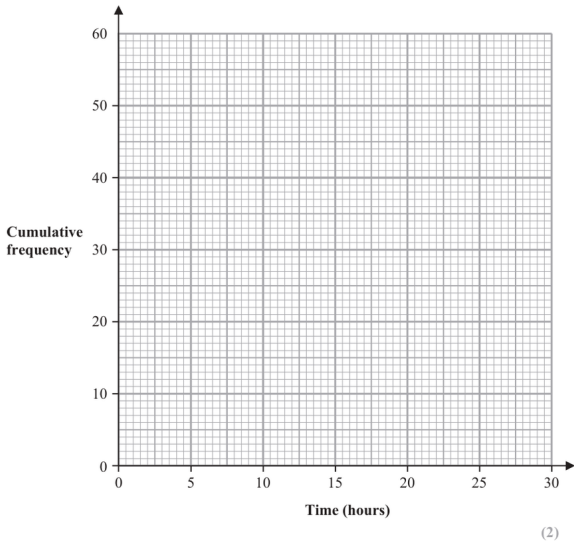
TOPIC: Cumulative Frequency Diagrams

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11 The cumulative frequency table gives information about the time, in hours, that each of 60 workers spent working from home in one week.

Time (t hours)	Cumulative frequency
$0 < t \leq 5$	6
$0 < t \leq 10$	17
$0 < t \leq 15$	27
$0 < t \leq 20$	42
$0 < t \leq 25$	53
$0 < t \leq 30$	60

(a) On the grid below, draw a cumulative frequency graph for the information in the table.



(2)

(b) Use your graph to find an estimate for the interquartile range of the times.

_____ hours
(2)

25 workers spent more than W hours working from home.

(c) Use your graph to find an estimate for the value of W

$W =$ _____
(2)

One of the 60 workers is chosen at random.
This worker spent H hours working from home.

(d) Find the probability that $5 < H \leq 10$

(1)

(Total for Question 11 is 7 marks)

PAST PAPER SOLUTION

Q#: 11

Marks: 7

TOPIC: **Cumulative Frequency Diagrams**

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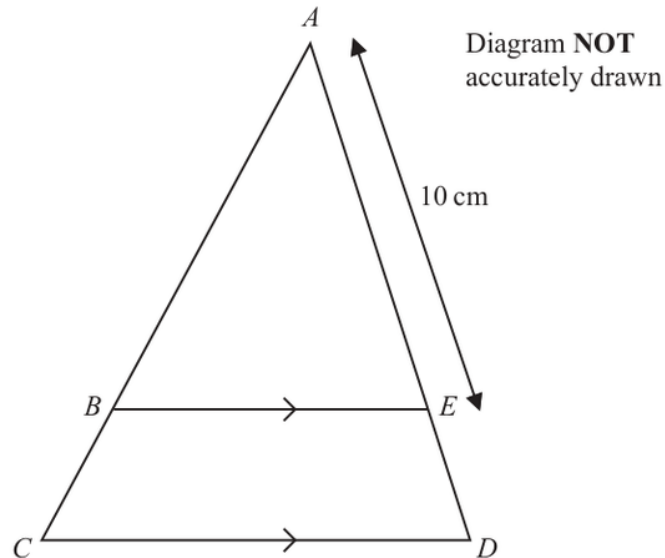
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EA

PAST PAPER QUESTION**Q#: 12****Marks: 4****TOPIC: Congruence, Similarity & Geometrical Proof**

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12

In the diagram, ABC and AED are straight lines.
 BE is parallel to CD

$$AE = 10 \text{ cm} \quad \text{and} \quad CD = 1.5 \times BE$$

(a) Work out the length of ED

..... cm
 (2)

$$AB = (2x + 5) \text{ cm and } BC = (3x - 5) \text{ cm}$$

(b) Work out the value of x

$x =$
 (2)

(Total for Question 12 is 4 marks)

PAST PAPER SOLUTION

Q#: 12

Marks: 4

TOPIC: **Congruence, Similarity & Geometrical Proof**

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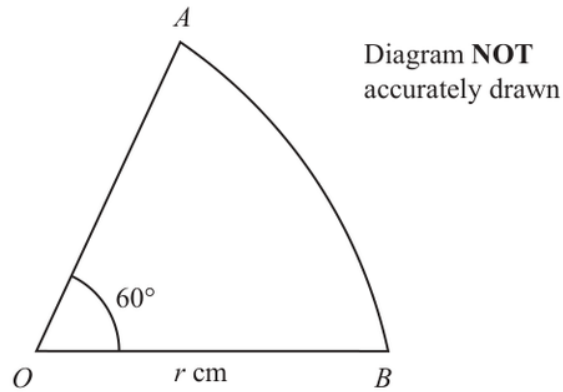
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EA

PAST PAPER QUESTION**Q#: 13****Marks: 3****TOPIC: Circles, Arcs & Sectors**

Jun 2024 P2HR - Jun 2024 | P2HR

13 OAB is a sector of a circle with centre O and radius r cm.



Angle $AOB = 60^\circ$

The perimeter of the sector is P cm.

Find a formula for P in terms of r

Give your answer in the form $P = r(c\pi + k)$ where c and k are values to be found.

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION

Q#: 13

Marks: 3

TOPIC: **Circles, Arcs & Sectors**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 14****Marks: 3****TOPIC: Probability Toolkit**

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14 Adriana is going to roll a biased dice and spin a biased coin.

The probability that the coin will land on Heads is 0.8

The probability that the dice will land on 6 and the coin will land on Heads is 0.24

Work out the probability that the dice will land on 6 and the coin will land on Tails.

(Total for Question 14 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 14

Marks: 3

TOPIC: **Probability Toolkit**

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WORKED SOLUTION

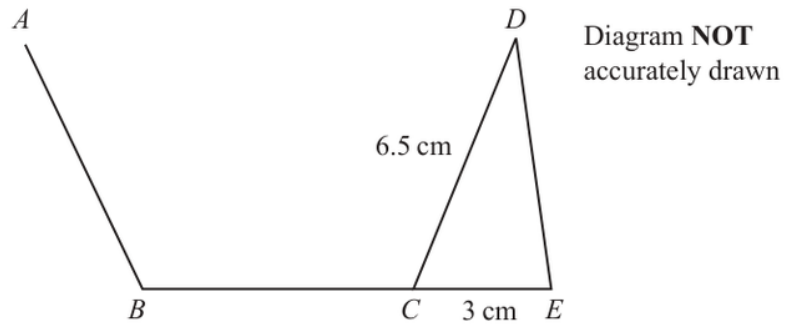
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PAST PAPER QUESTION**Q#: 15****Marks: 3****TOPIC: Sine, Cosine Rule & Area of Triangles**

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15

AB , BC and CD are three sides of a regular pentagon and CDE is a triangle.
 BCE is a straight line.

$$CD = 6.5 \text{ cm} \quad CE = 3 \text{ cm}$$

Work out the area of triangle CDE
 Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION

Q#: 15

Marks: 3

TOPIC: **Sine, Cosine Rule & Area of Triangles**

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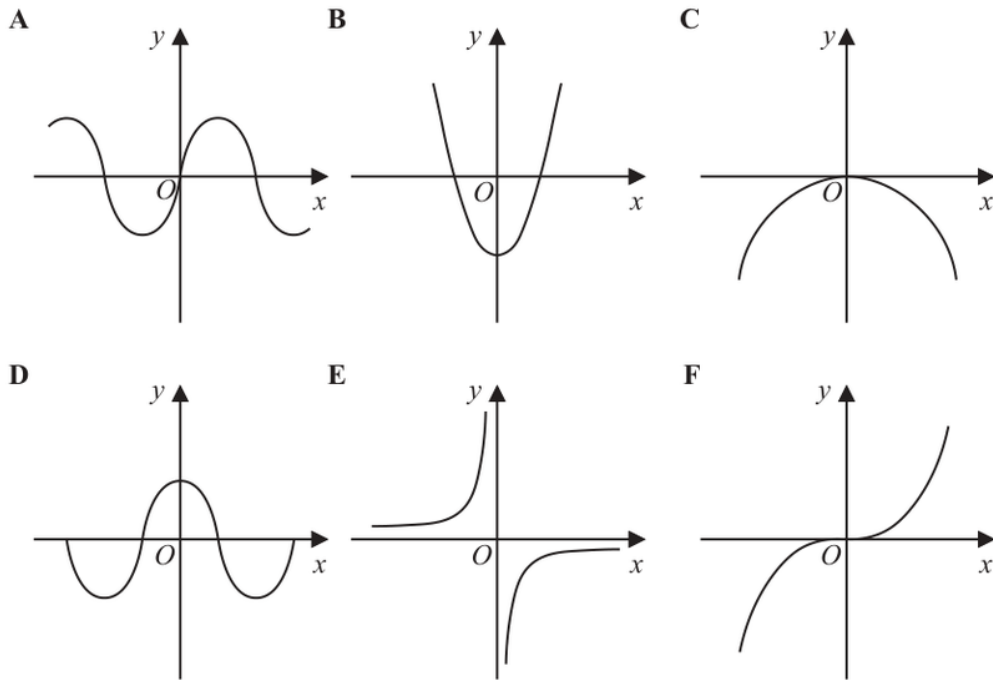
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EA

PAST PAPER QUESTION**Q#: 16****Marks: 2****TOPIC: Graphs of Functions**

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16 Here are six graphs.

Write down the letter of the graph that could have the equation

(i) $y = -\frac{1}{x}$

.....
(1)

(ii) $y = \sin x^\circ$

.....
(1)**(Total for Question 16 is 2 marks)**

PAST PAPER SOLUTION

Q#: 16

Marks: 2

TOPIC: **Graphs of Functions**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 17****Marks: 3****TOPIC: Functions**

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17 $f(x) = \frac{x}{2x-4}$ $g(x) = 3x + 1$

Given that $fg(k) = 2$

work out the value of k

$k = \dots\dots\dots$

(Total for Question 17 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 17

Marks: 3

TOPIC: **Functions**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 18****Marks: 2**TOPIC: **Algebraic Proof**

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18 Use algebra to show that $0.\dot{3}0\dot{6} = \frac{34}{111}$

(Total for Question 18 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 18

Marks: 2

TOPIC: **Algebraic Proof**

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EA

PAST PAPER QUESTION**Q#: 19****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

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19 Aviv goes on a cycle journey.

For the cycle journey

average speed = 19 km/h correct to the nearest whole number

time = 1.5 hours correct to one decimal place

Work out the upper bound for the distance Aviv travels.

Give your answer correct to 3 significant figures.

..... km

(Total for Question 19 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 19

Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Solving Inequalities**

Jun 2024 P2HR - Jun 2024 | P2HR

- 20** Solve $6x^2 - 7x - 20 > 0$
Show clear algebraic working.

(Total for Question 20 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 20

Marks: 4

TOPIC: **Solving Inequalities**

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WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Solving Inequalities**

Jun 2024 P2HR - Jun 2024 | P2HR

- 20** Solve $6x^2 - 7x - 20 > 0$
Show clear algebraic working.

(Total for Question 20 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 20

Marks: 4

TOPIC: **Solving Inequalities**

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EA

PAST PAPER QUESTION**Q#: 21****Marks: 4****TOPIC: Coordinate Geometry**

Jun 2024 P2HR - Jun 2024 | P2HR

21 $ABCD$ is a square.The point A has coordinates $(-5, 2)$ The point B has coordinates $(3, 5)$ Find an equation of the line that passes through B and C Give your answer in the form $ax + by + c = 0$ where a , b and c are integers.

(Total for Question 21 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 21

Marks: 4

TOPIC: **Coordinate Geometry**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 21****Marks: 4****TOPIC: Coordinate Geometry**

Jun 2024 P2HR - Jun 2024 | P2HR

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(Total for Question 21 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 21

Marks: 4

TOPIC: **Coordinate Geometry**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 22****Marks: 5****TOPIC: Simultaneous Equations**

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22 Solve the simultaneous equations

$$x^2 + y^2 = y + 11$$

$$y = 3x - 1$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 22

Marks: 5

TOPIC: **Simultaneous Equations**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 22****Marks: 5****TOPIC: Simultaneous Equations**

Jun 2024 P2HR - Jun 2024 | P2HR

22 Solve the simultaneous equations

$$x^2 + y^2 = y + 11$$

$$y = 3x - 1$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 22

Marks: 5

TOPIC: **Simultaneous Equations**

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PAST PAPER QUESTION**Q#: 23****Marks: 2****TOPIC: Transformations of Graphs**

Jun 2024 P2HR - Jun 2024 | P2HR

23 A curve has equation $y = f(x)$ The coordinates of the minimum point on this curve are $(6, -3)$

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x) + 10$

(.....,)
(1)

(ii) $y = f(3x)$

(.....,)
(1)

(Total for Question 23 is 2 marks)

PAST PAPER SOLUTION

Q#: 23

Marks: 2

TOPIC: **Transformations of Graphs**

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Jun 2024 P2HR - Jun 2024 | P2HR

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(..... ,)
(1)

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(..... ,)
(1)

(Total for Question 23 is 2 marks)

PAST PAPER SOLUTION

Q#: 23

Marks: 2

TOPIC: **Transformations of Graphs**

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PAST PAPER QUESTION**Q#: 24****Marks: 5****TOPIC: Area & Volume of Similar Shapes**

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24 The diagram shows a solid, **S**, made from a cone and a hemisphere.

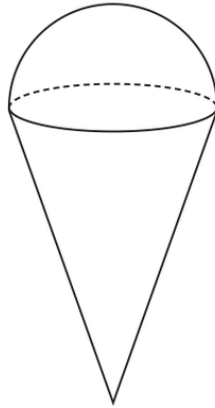


Diagram **NOT**
accurately drawn

The centre of the circular face of the cone coincides with the centre of the flat surface of the hemisphere.

The radius of the circular face of the cone, x cm, is equal to the radius of the hemisphere.

The total height of **S** is $4 \times$ the radius of the hemisphere.

A separate sphere has radius kx cm.

The volume of this sphere is $12.5 \times$ the volume of **S**

(a) Work out the value of k

$$k = \dots\dots\dots (4)$$

A solid, **T**, is similar to solid **S**

The volume of **T** is $512 \times$ the volume of **S**

The total surface area of **T** is $d \times$ the total surface area of **S**

(b) Find the value of d

$$d = \dots\dots\dots (1)$$

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24

Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

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PAST PAPER QUESTION**Q#: 24****Marks: 5****TOPIC: Area & Volume of Similar Shapes**

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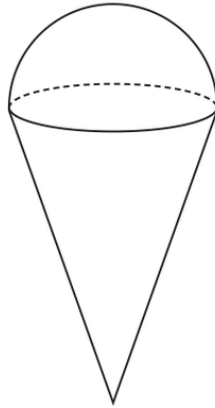


Diagram **NOT**
accurately drawn

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(b) Find the value of d

$$d = \dots\dots\dots (1)$$

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24

Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

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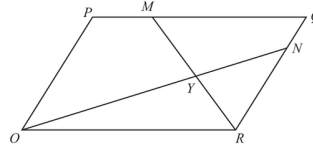
PAST PAPER QUESTION

Q#: 25

Marks: 6

TOPIC: **Vectors**

Jun 2024 P2HR - Jun 2024 | P2HR

25 $OPQR$ is a parallelogram.Diagram NOT
accurately drawn

$$\vec{OP} = 2\mathbf{a} \quad \text{and} \quad \vec{OR} = 3\mathbf{b}$$

The point M lies on PQ such that $PM = \frac{1}{4}PQ$

The point N lies on RQ such that $RN = \frac{4}{5}RQ$

(a) Find, in terms of $\mathbf{a}$ and $\mathbf{b}$, giving your answers in simplest form

(i) $\vec{ON}$

(1)

(ii) $\vec{MR}$

(1)

MR and ON intersect at the point Y

Given that

$$OY = k \times ON$$

(b) use a vector method to find the value of k

$k =$ _____
(4)

(Total for Question 25 is 6 marks)

PAST PAPER SOLUTION

Q#: 25

Marks: 6

TOPIC: **Vectors**

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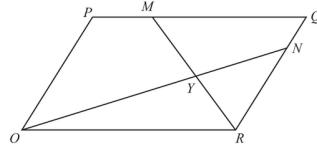
PAST PAPER QUESTION

Q#: 25

Marks: 6

TOPIC: **Vectors**

Jun 2024 P2HR - Jun 2024 | P2HR

25 $OPQR$ is a parallelogram.Diagram NOT
accurately drawn

$$\vec{OP} = 2\mathbf{a} \quad \text{and} \quad \vec{OR} = 3\mathbf{b}$$

The point M lies on PQ such that $PM = \frac{1}{4}PQ$

The point N lies on RQ such that $RN = \frac{4}{5}RQ$

(a) Find, in terms of $\mathbf{a}$ and $\mathbf{b}$, giving your answers in simplest form

(i) $\vec{ON}$

(1)

(ii) $\vec{MR}$

(1)

MR and ON intersect at the point Y

Given that

$$OY = k \times ON$$

(b) use a vector method to find the value of k

$$k = \underline{\hspace{2cm}}$$

(4)

(Total for Question 25 is 6 marks)

PAST PAPER SOLUTION

Q#: 25

Marks: 6

TOPIC: **Vectors**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 26****Marks: 4****TOPIC: Algebraic Fractions**

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26 Write $4 - \left[(3x - 5) \div \frac{3x^2 + x - 10}{4x - 1} \right]$ as a single fraction in its simplest form.

(Total for Question 26 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 26

Marks: 4

TOPIC: **Algebraic Fractions**

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WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 26****Marks: 4****TOPIC: Algebraic Fractions**

Jun 2024 P2HR - Jun 2024 | P2HR

26 Write $4 - \left[(3x - 5) \div \frac{3x^2 + x - 10}{4x - 1} \right]$ as a single fraction in its simplest form.

(Total for Question 26 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 26

Marks: 4

TOPIC: **Algebraic Fractions**

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WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA